



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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SAVEETHA SCHOOL OF ENGINEERING, SIMATS
Department of Computer Science and Engineering
Assignment Information

Particular	Details
Name:	G. Venkata Niranjan Reddy
Register Number:	192525199
Department:	AI & ML
Programme:	BTECH
Course Code & Course Name	CSA1522 – Cloud Computing and Big Data Analytics
Academic Year / Batch	2026–27
Faculty Name	Dr P Rajaram
Assignment Title	Cloud-Based Placement Data Management and Big Data Analytics using JOHO
Date of Issue	31-8-2026
Date of Submission	01-9-2026
Maximum Marks	100
Course Outcome(s) – CO	CO1: Understand the cloud services with different service models. CO2: Demonstrate the deployment of Virtualization and build a data center using virtualization tools. CO3: Build methods to access cloud and various techniques to collaborate within cloud.
Bloom's Taxonomy Level	L3 – Apply and L4 – Analyse
SDG Mapping	SDG 4 – Quality Education SDG 8 – Decent Work and Economic Growth SDG 9 – Industry Innovation and Infrastructure SDG 11 – Sustainable Cities and Communities
Industry	Applies cloud services, virtualization and distributed big-data processing to a JOHO-based placement workflow, supporting scalable access, collaboration and analytics while considering security, privacy and responsible data use.

Problem Statement:

Design and evaluate a simple cloud-based placement data management and access solution for a university using the JOHO platform. The placement cell needs a reliable way to organize placement information and provide controlled access to students and placement staff. Students shall select suitable IaaS, PaaS and SaaS service models, design a basic cloud architecture, create and configure virtual machines using Oracle VM VirtualBox, demonstrate secure access to a cloud-hosted web application or Web API, and show how users can collaborate through shared cloud resources.

Assessment Rubrics

Assessment Criterion / Marks	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
CO1 – Cloud Service Models & Architecture 25 Marks	Correctly identifies the JOHO requirements, selects appropriate IaaS/PaaS/SaaS models, and presents a complete architecture with clear justification.	Selects appropriate service models and presents a suitable architecture with minor gaps in justification or detail.	Identifies basic service models and provides a simple architecture, but the justification or mapping to requirements is limited.	Service models are unsuitable or misunderstood; architecture is missing/unclear and justification is inadequate.
CO2 – Virtualization & VM Deployment 25 Marks	Creates and configures two VMs correctly using Oracle VM VirtualBox; records CPU, memory, storage and networking; demonstrates communication and clearly explains the hypervisor.	Creates and configures both VMs successfully with mostly correct resources and networking; communication works with minor issues.	Creates the VMs and demonstrates basic operation, but configuration, networking, testing or explanation is incomplete.	VM creation/configuration is incorrect or incomplete; communication is not demonstrated and supporting evidence is insufficient.
CO3 – Cloud Access & Collaboration 25 Marks	Demonstrates secure cloud access successfully, including authentication/access control, clear request-response evidence and a practical collaboration workflow.	Demonstrates working cloud access and collaboration with minor gaps in security, explanation or evidence.	Demonstrates basic cloud access, but collaboration, authentication or security evidence is limited.	Cloud access or collaboration is non-functional, unclear, insecure or not supported by evidence.
Analysis, Validation & Engineering Justification 15 Marks	Compares feasible alternatives using clear criteria, performs relevant tests, interprets results accurately and gives a well-supported final decision.	Provides a reasonable comparison and testing with generally correct interpretation and justification.	Provides limited comparison or testing; interpretation and justification are basic.	Little or no meaningful comparison, testing, interpretation or engineering justification is provided.
Security, Privacy, Ethics & SDG Relevance 10 Marks	Clearly addresses security, privacy, ethical handling of placement data and relevant SDGs with specific application to the JOHO scenario.	Addresses relevant security, privacy, ethical and SDG considerations with minor omissions.	Mentions general security, privacy, ethics or SDG points but gives limited connection to the problem.	Important security, privacy, ethical or SDG considerations are missing or inappropriate.

Deliverables

To receive full credit, each submission must include the following deliverables:

1. Pseudo code
2. Implementation & Results
3. Github and GCR Upload

Assignment Problem / Challenge

Problem / Case

A university placement cell currently maintains student placement information using spreadsheets, local computer systems and manually shared documents. This approach can result in:

- Duplicate student records
- Difficulty in accessing placement information
- Unauthorized access to student data
- Lack of centralized data management
- Difficulty in updating placement status
- Limited collaboration between students and placement staff
- Dependency on a particular computer or local system

The university wants to develop a **simple cloud-based placement data management and access solution using the Zoho platform.**

Students are required to design and evaluate a basic cloud solution that demonstrates:

1. Selection of appropriate **IaaS, PaaS and SaaS** service models.
2. A basic **cloud architecture** for the placement system.
3. Creation and configuration of **virtual machines using Oracle VM VirtualBox.**
4. Secure access to a **cloud-hosted web application or Web API.**
5. Controlled access for different categories of users.
6. Collaboration using shared cloud resources.
7. Testing and evaluation of the proposed solution.

The solution should demonstrate how cloud computing can improve the **availability, accessibility, security and collaboration** of placement data.

C. Problem Statement

Design Challenge

Design and evaluate a simple cloud-based placement data management and access solution for a university using the Zoho platform.

The proposed system should allow placement staff to manage information such as:

- Student ID
- Student name

- Department
- Email
- CGPA
- Skills
- Eligible companies
- Company name
- Job role
- Placement status
- Package
- Interview status

The system should provide **controlled access** to different users.

Suggested User Roles

User	Access
Administrator	Full system access
Placement Staff	Add, update and manage placement records
Student	View/update permitted personal information and placement status
Company/Recruiter	View only the information authorized by the placement cell

D. Requirements and Constraints

The assignment format requires relevant functional, performance, technical, security/privacy, cost and user requirements to be considered where applicable.

1. Functional Requirements

The proposed system should:

1. Store student placement information centrally.
2. Allow placement staff to add student records.
3. Allow authorized staff to update placement information.
4. Allow students to access their permitted information.
5. Provide company/placement information.
6. Maintain placement status.

7. Provide search/filter functionality.
8. Support controlled access.
9. Demonstrate cloud-based collaboration.
10. Provide backup or export of important data.

2. Technical Requirements

Students should demonstrate:

- Zoho platform usage
- Cloud service models
- Basic cloud architecture
- Oracle VM VirtualBox
- Virtual machine creation
- Web application/Web API access
- Authentication
- Role-based access
- Shared cloud resources

3. Security Requirements

The solution should consider:

- User authentication
- Authorization
- Role-based access
- Password protection
- Secure communication
- Protection of student personal information
- Restricted access to sensitive placement information

4. Performance Requirements

The system should provide:

- Reasonable response time
- Reliable access
- Centralized data availability

- Ability to handle multiple student records
- Consistent data updates

5. Cost Constraints

The proposed solution should preferably use:

- Free/student-accessible cloud services where possible
- Open-source or freely available virtualization software
- Minimum infrastructure requirements

6. Other Constraints

- Student data privacy must be maintained.
- Only authorized users should access sensitive information.
- The solution should be simple enough to demonstrate in a laboratory environment.
- Screenshots/evidence of tool usage should be provided.

E. Student Work

The PDF requires students to explain the problem, expected outcomes, available data, assumptions and constraints before developing the solution.

1. Problem Understanding and Formulation

Problem

The existing manual placement data management approach makes it difficult to maintain a centralized and secure placement database.

Expected Outcome

The expected outcome is a **cloud-based placement management system** that:

- Centralizes placement information.
- Provides controlled access.
- Improves accessibility.
- Supports collaboration.
- Demonstrates virtualization.
- Provides secure web/API access.
- Reduces dependency on local systems.

Available Data

Sample data can contain:

Student ID	Name	Department	CGPA	Skills	Company	Status
ST001	Student A	CSE	8.7	Python, SQL	ABC Ltd	Selected
ST002	Student B	CSE	8.2	Java, SQL	XYZ Ltd	Interview
ST003	Student C	ECE	7.9	Python, IoT	PQR Ltd	Applied

Students may use **synthetic/sample data** rather than real personal information.

Assumptions

- Users have internet access.
- The university has authorized placement staff.
- Zoho services are available.
- Student information can be represented using sample records.
- Oracle VM VirtualBox is available on the student's computer.

2. Application of Course Knowledge

Students should apply the following cloud computing concepts.

A. IaaS – Infrastructure as a Service

Example: Virtual machine infrastructure.

The student can demonstrate IaaS concepts by creating a VM using **Oracle VM VirtualBox**.

Example:

Host Computer → VirtualBox → Linux/Windows VM → Web Server/API

The VM demonstrates the concept of providing computing infrastructure on demand.

B. PaaS – Platform as a Service

The application development/deployment environment can be represented through the platform capabilities used to develop and expose the placement application/API.

C. SaaS – Software as a Service

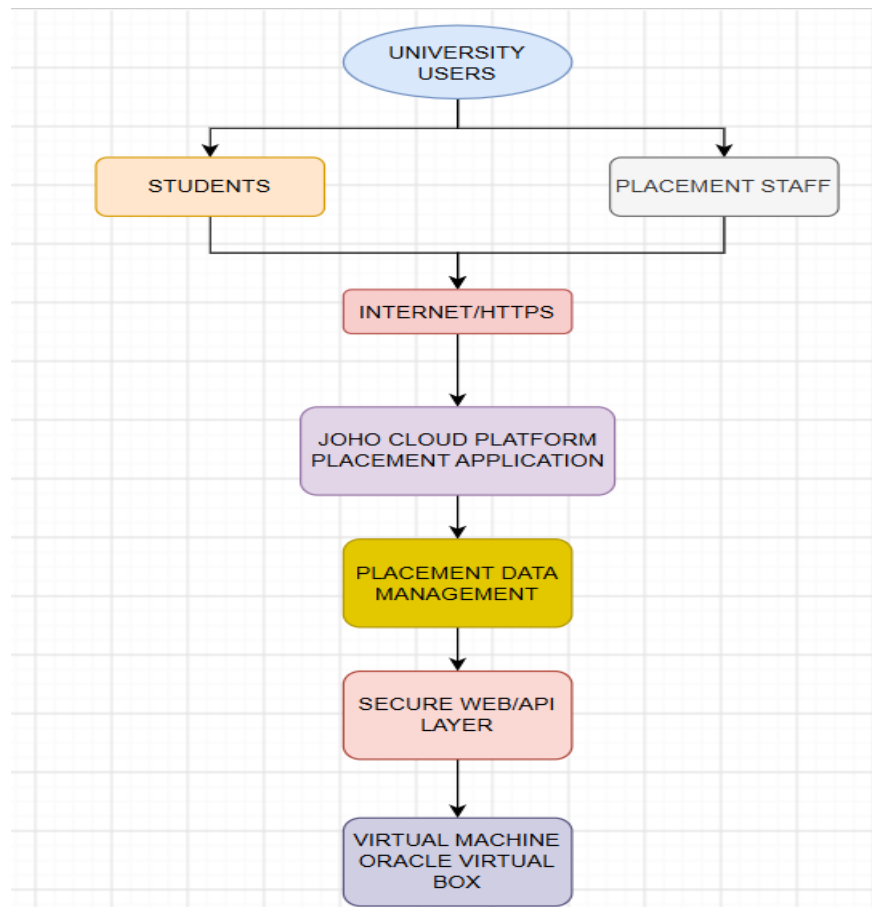
The Zoho-based placement management application can be considered the SaaS-facing component because users access the application through a web interface without managing the underlying infrastructure.

Service Model Comparison

Feature	IaaS	PaaS	SaaS
Main purpose	Infrastructure	Application development/deployment	Ready-to-use application
Example in assignment	Virtual Machine	Application/API platform	Zoho application
User management	High	Medium	Low
Infrastructure management	User	Provider	Provider
Example task	Create VM	Deploy application/API	Manage placement records

3. Proposed Cloud Architecture

Basic Architecture



4. Solution Methodology / Implementation

The PDF specifies that the solution may include architecture, process, prototype, simulation, case analysis and other implementation evidence.

Step 1 – Design the Placement Data Structure

Create the following basic entities:

Student

- Student_ID
- Name
- Department
- Email
- CGPA
- Skills
- Graduation Year

Company

- Company_ID
- Company_Name
- Job_Role
- Required_Skills
- Minimum_CGPA
- Package

Placement

- Placement_ID
- Student_ID
- Company_ID
- Interview_Status
- Placement_Status
- Package

Step 2 – Create the Zoho Application

Create a basic placement management application using the Zoho platform.

Suggested modules/forms:

1. **Student Registration**
2. **Company Details**
3. **Job/Placement Drive**
4. **Student Applications**
5. **Interview Status**
6. **Final Placement Status**

Step 3 – Configure User Access

Implement different access permissions.

Administrator

- Add users
- View all records
- Modify all records
- Manage application

Placement Staff

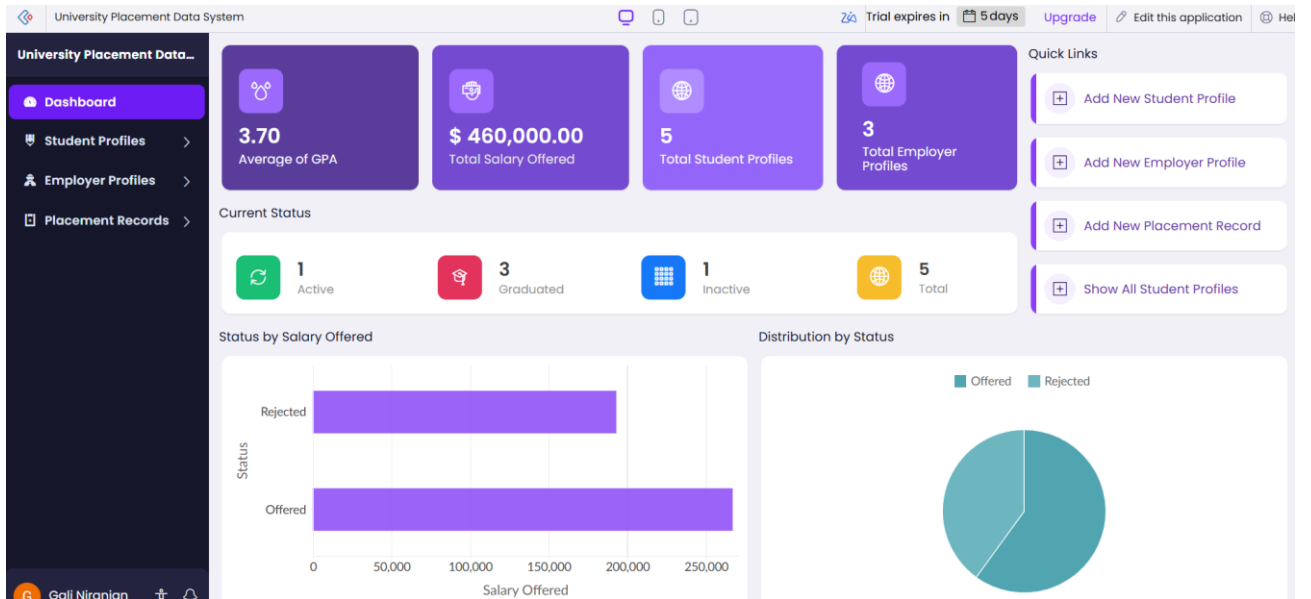
- Add student records
- Add company information
- Update placement status
- Generate reports

Student

- View personal details
- View eligible drives
- Apply for permitted opportunities
- View application/placement status

Recruiter

- View authorized candidate information
- View relevant applications



University Placement Data...

- Dashboard
- Student Profiles
- + Student Profile
- All Student Profiles
- Current Statuses
- Employer Profiles
- Placement Records

Student Profile

Student Id

Full Name

First Name

Email Address

Phone Number

+91

81234 56789

Major

Graduation Year

#####

GPA

#####

Resume URL

https://

LinkedIn Profile

https://

Current Status

-Select-

University Placement Data...

- Dashboard
- Student Profiles
- Employer Profiles
- + Employer Profile
- All Employer Profiles
- Company Sizes
- Placement Records

Employer Profile

Company Code

Company Name

Industry

Headquarters

Address Line 1

Website

https://

Contact Email

Contact Phone

+91

81234 56789

Recruiter Name

First Name

Recruiter Title

Company Size

-Select-

University Placement Data... Dashboard Student Profiles > Employer Profiles > Placement Records ▾ Placement Record All Placement Reco... Employment Types	All Placement Records									
	☐	Record Id ▾	Offer Date ▾	Job Title ▾	Salary Offered ▾	Joining Date ▾	Employment Type ▾	Benefits ▾	Status ▾	Notes
		1005	19-Aug-2026	DevOps Engineer	\$ 98,000.00	15-Sep-2026	Full-time	Relocation Aid	Rejected	Full tin
		1004	18-Aug-2026	UX Designer	\$ 85,000.00	12-Sep-2026	Part-time	Gym Membership	Offered	Flexibl
		1003	17-Aug-2026	Product Manager	\$ 110,000.00	10-Sep-2026	Contract	Stock Options	Offered	Hybrid
		1002	16-Aug-2026	Data Analyst	\$ 72,000.00	05-Sep-2026	Part-time	401k Plan	Offered	On-sit
		1001	15-Aug-2026	Senior Software Engineer	\$ 95,000.00	01-Sep-2026	Contract	Health Insurance	Rejected	Remot availa

5. Oracle VM VirtualBox Demonstration

Students should create a virtual machine using **Oracle VM VirtualBox**.

Suggested Configuration

Parameter	Suggested Value
VM Name	Placement-Server
Operating System	Ubuntu/Linux
RAM	2–4 GB
CPU	2 cores
Storage	20 GB
Network	NAT / Bridged Adapter

Implementation Steps

Step 1: Open Oracle VM VirtualBox.

Step 2: Select New.

Step 3: Enter VM name as Placement_Server.

Step 4: Select the required operating system.

Step 5: Allocate approximately 2 GB RAM.

Step 6: Allocate 20 GB virtual storage.

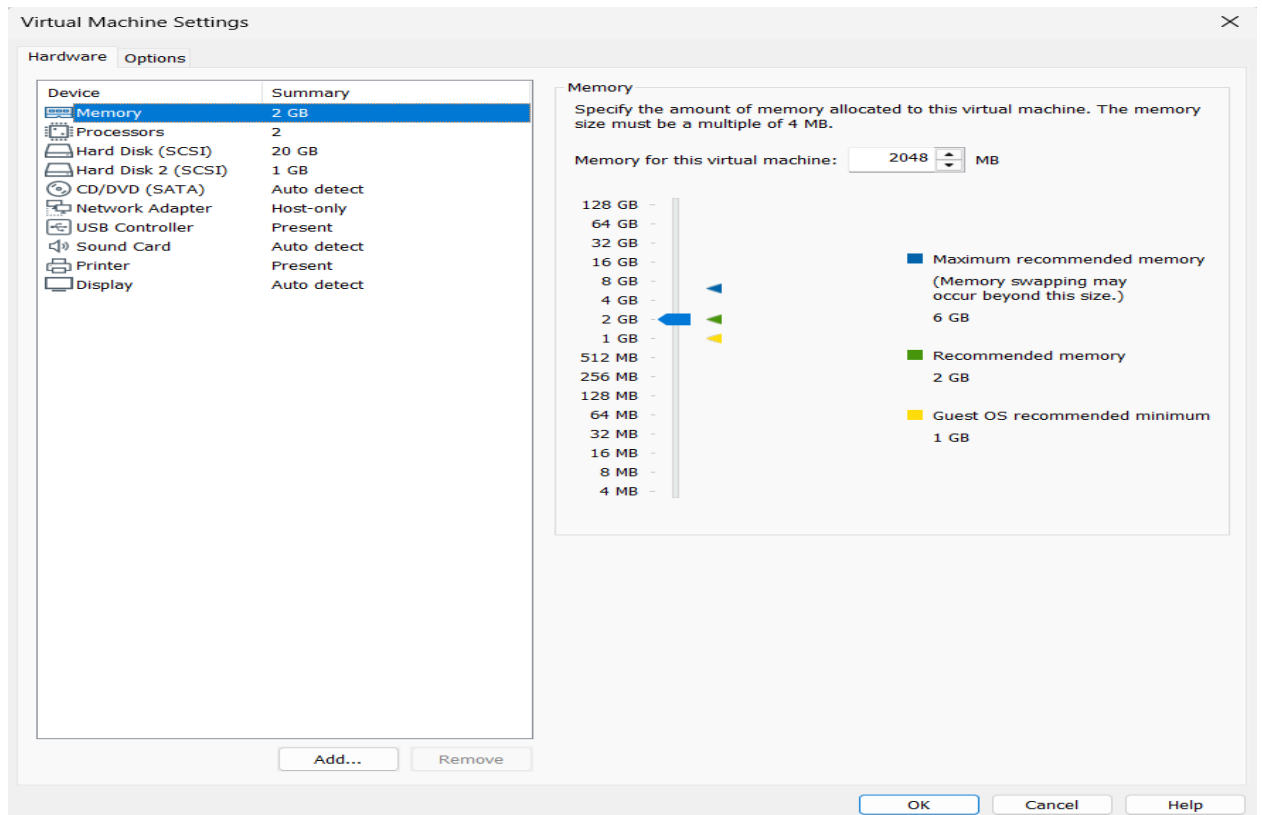
Step 7: Configure network connectivity.

Step 8: Start the virtual machine.

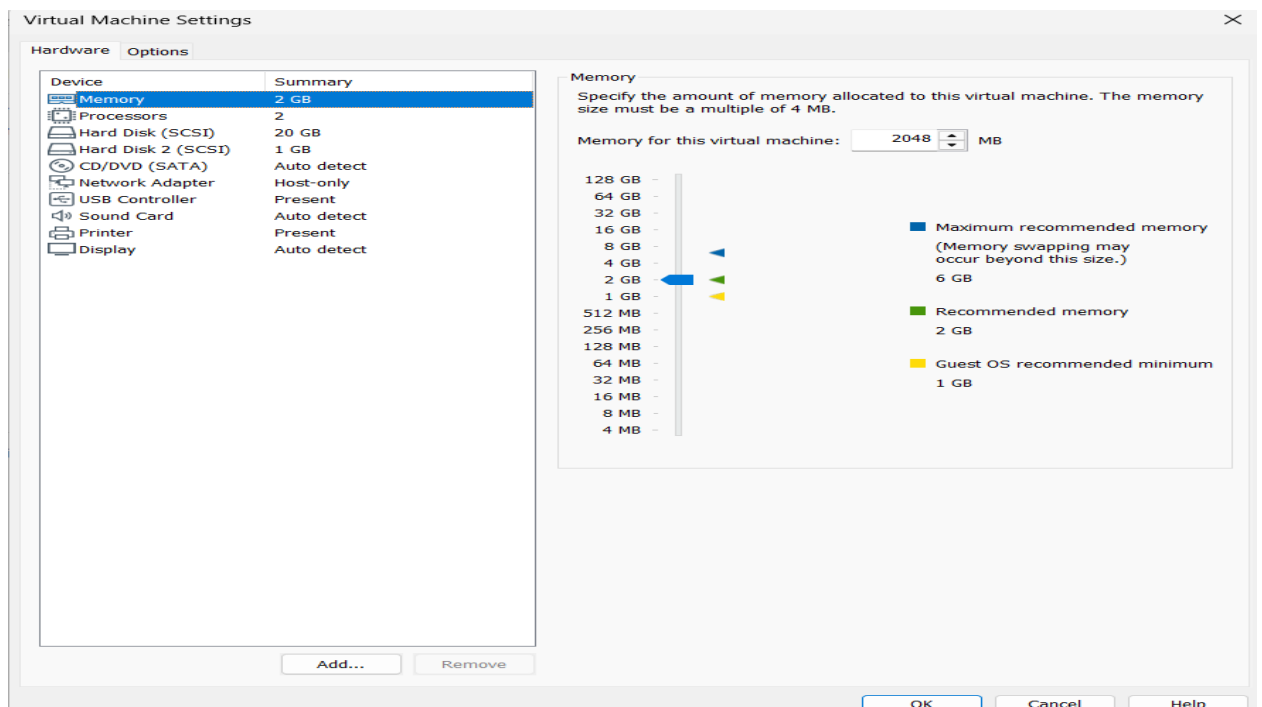
Step 9: Install the operating system.

Step 10: Configure the environment required for the Web application/API.

1. VMware Placement-Server configuration.



2. VMware Placement-Client configuration.



3. Placement-Server IP address.

```

File Edit View Search Terminal Help
    valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
    valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
    link/ether 00:0c:29:7d:3c:87 brd ff:ff:ff:ff:ff:ff
    inet 192.168.204.128/24 brd 192.168.204.255 scope global dynamic noprefixrou
te ens33
    valid_lft 1747sec preferred_lft 1747sec
    inet6 fe80::666c:9e7e:7e33:d290/64 scope link noprefixroute
    valid_lft forever preferred_lft forever

```

4. Placement-ClientIPAddress.

```

File Edit View Search Terminal Help
dhilip@Placement-Client:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
    link/ether 00:0c:29:5c:07:f3 brd ff:ff:ff:ff:ff:ff
    inet 192.168.204.129/24 brd 192.168.204.255 scope global dynamic noprefixrou
te ens33
    valid_lft 1732sec preferred_lft 1732sec
    inet6 fe80::4629:c605:110a:572a/64 scope link noprefixroute
    valid_lft forever preferred_lft forever

```

5. Client-to-Server successful ping.

```

dhilip@Placement-Server:~$ ping -c 4 192.168.204.129
PING 192.168.204.129 (192.168.204.129) 56(84) bytes of data.
64 bytes from 192.168.204.129: icmp_seq=1 ttl=64 time=4.13 ms
64 bytes from 192.168.204.129: icmp_seq=2 ttl=64 time=1.71 ms
64 bytes from 192.168.204.129: icmp_seq=3 ttl=64 time=3.06 ms
64 bytes from 192.168.204.129: icmp_seq=4 ttl=64 time=1.61 ms

--- 192.168.204.129 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 1.612/2.630/4.134/1.040 ms

```

6. Server-to-Client successful ping.

```

dhilip@Placement-Client:~$ ping -c 4 192.168.204.128
PING 192.168.204.128 (192.168.204.128) 56(84) bytes of data.
64 bytes from 192.168.204.128: icmp_seq=1 ttl=64 time=4.31 ms
64 bytes from 192.168.204.128: icmp_seq=2 ttl=64 time=1.81 ms
64 bytes from 192.168.204.128: icmp_seq=3 ttl=64 time=2.24 ms
64 bytes from 192.168.204.128: icmp_seq=4 ttl=64 time=1.08 ms

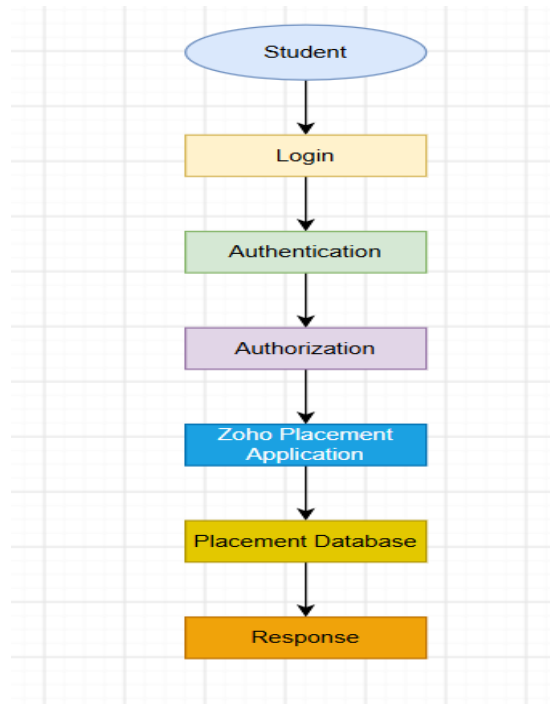
--- 192.168.204.128 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.085/2.362/4.312/1.199 ms

```

6. Secure Web Application / Web API Demonstration

Students should demonstrate access to a cloud-hosted web application or Web API.

Example Flow



Security Demonstration

Students should demonstrate:

- Login authentication
- User roles
- Authorized/unauthorized access
- HTTPS/secure access where supported
- Restricted placement information
- Logout/session control

Test Case	Expected Result
Correct username/password	Login successful
Incorrect password	Login rejected
Student accesses own record	Access granted
Student accesses another student's private data	Access denied
Student accesses admin page	Access denied
Placement staff updates placement status	Successful

Administrator accesses all records	Successful
Unauthorized API request	Rejected

7. Cloud Collaboration Demonstration

Students should demonstrate how multiple users can collaborate through shared cloud resources.

Examples of collaboration:

- Placement staff updates company information.
- Students view available placement drives.
- Students update permitted information.
- Placement staff updates interview results.
- Administrators monitor placement statistics.
- Authorized users access updated information from different devices.

8. Use of Modern Tools

The assignment format specifically asks students to provide appropriate evidence of modern-tool usage and outputs.

Tools

Tool	Purpose
Zoho Platform	Cloud-based placement application
Oracle VM VirtualBox	Virtual machine creation
Web Browser	Access cloud application
Web/API Tools	Test application/API
Linux/Windows VM	Server/application environment
Spreadsheet/CSV	Sample placement dataset
Cloud Storage	Shared resources/backup

9. Results and Validation

The PDF recommends presenting results through tables, graphs, screenshots, performance metrics and test results, followed by validation against requirements.

Functional Test Table

Test Case	Expected Result	Actual Result	Status
Admin login	Login successful	Successful	Pass
Student login	Student dashboard displayed	Displayed	Pass
Invalid login	Access denied	Access denied	Pass
Add student	Record created	Created	Pass
Update placement status	Status updated	Updated	Pass
Unauthorized admin access	Access denied	Denied	Pass
Shared data access	Authorized users can view	Successful	Pass
VM startup	VM boots correctly	Successful	Pass
Web/API request	Valid response	Received	Pass

Requirement Validation

Requirement	Validation Method	Result
Centralized data	Application database	Satisfied
Controlled access	Role testing	Satisfied
Virtualization	VirtualBox VM	Satisfied
Cloud application	Zoho application	Satisfied
Secure access	Authentication/authorization test	Satisfied
Collaboration	Multiple-user demonstration	Satisfied
Data management	CRUD testing	Satisfied

10. Analysis and Engineering Decision

Local System vs Cloud-Based System

Factor	Local System	Proposed Cloud System
Data location	Local computer	Cloud
Accessibility	Limited	Remote access
Collaboration	Difficult	Easier
Centralized management	Limited	Available
Scalability	Limited	Better
Maintenance	Local maintenance	Platform-supported
User access control	Can be limited	Role-based access
Data sharing	Manual	Cloud-based
Dependency	Local infrastructure	Internet/cloud service

Advantages

- Centralized data management
- Improved accessibility
- Better collaboration
- Controlled user access
- Easier information sharing
- Reduced duplicate files
- Demonstrates cloud computing concepts

Limitations

- Requires internet connectivity.
- Depends on cloud-platform availability.
- Student data must be carefully protected.
- Free-tier/cloud-platform limitations may exist.
- Virtual machine performance depends on host computer resources.

Final Engineering Decision

The cloud-based solution is selected because it provides a better environment for centralized placement-data management and controlled access than a purely local system.

11. Broader Considerations

Security and Privacy

Student placement information can contain sensitive personal information. Therefore:

- Only authorized users should access records.
- Students should not see another student's private information.
- Passwords should not be shared.
- Sample data should preferably be used for demonstrations.

Economic Consideration

Using cloud services can reduce the need for dedicated university hardware and simplify maintenance.

Sustainability

Cloud-based access can reduce duplication of locally stored files and physical infrastructure requirements.

Accessibility

Students and placement staff can potentially access placement information from different locations and devices through the web.

Professional Responsibility

The placement cell must ensure that student information is used only for legitimate academic and recruitment purposes.

12. Conclusion

The proposed project presents a simple cloud-based placement data management and access solution using the JOHO platform. The system centralizes student, company, placement-drive, application and selection information in a cloud-based environment.

The solution demonstrates the three cloud service models by associating virtualized infrastructure with IaaS, application development/deployment capabilities with PaaS and the JOHO-based application with SaaS. Oracle VM VirtualBox provides a practical demonstration of virtual machine creation and configuration.

Role-based access provides controlled access for administrators, placement staff, students and authorized recruiters. Secure Web/API access demonstrates how placement information can be accessed through an authenticated application.

The testing results demonstrate that the proposed design can satisfy the identified functional requirements. The major limitations are internet dependency, cloud-service dependency and the need for strong data privacy controls.

Future improvements can include automated email notifications, advanced placement analytics, mobile access, stronger authentication, automated backups and integration with the university student information system.

13. Student Reflection

1. What did I learn from this assignment?

I learned how cloud computing concepts can be applied to solve a real-world university placement management problem. I understood the differences between IaaS, PaaS and SaaS and how these models can be combined in a practical cloud solution.

2. What did I learn about virtualization?

I learned how Oracle VM VirtualBox can be used to create and configure a virtual machine with CPU, memory, storage and network resources.

3. What did I learn about security?

I learned that cloud applications require authentication and authorization to prevent unauthorized users from accessing confidential placement information.

4. What challenges did I face?

The main challenges were designing appropriate user roles, organizing placement data and understanding how the cloud application and virtual machine components can work together.

5. What would I improve?

If additional time and resources were available, I would add automated notifications, dashboards, analytics, stronger authentication, automated backups and integration with the university's existing student database.

The reflection requirements in the PDF specifically ask what was learned beyond classroom teaching and what would be improved with additional time/resources.

14. References

- Cloud Computing – course notes and prescribed textbook.
- JOHO Platform – official platform documentation.
- Oracle VM VirtualBox – official documentation.
- Relevant cloud computing research papers.
- Web/API documentation used during implementation.
- Software documentation for tools used in the project.